PEAD strategy suite -- comprehensive report

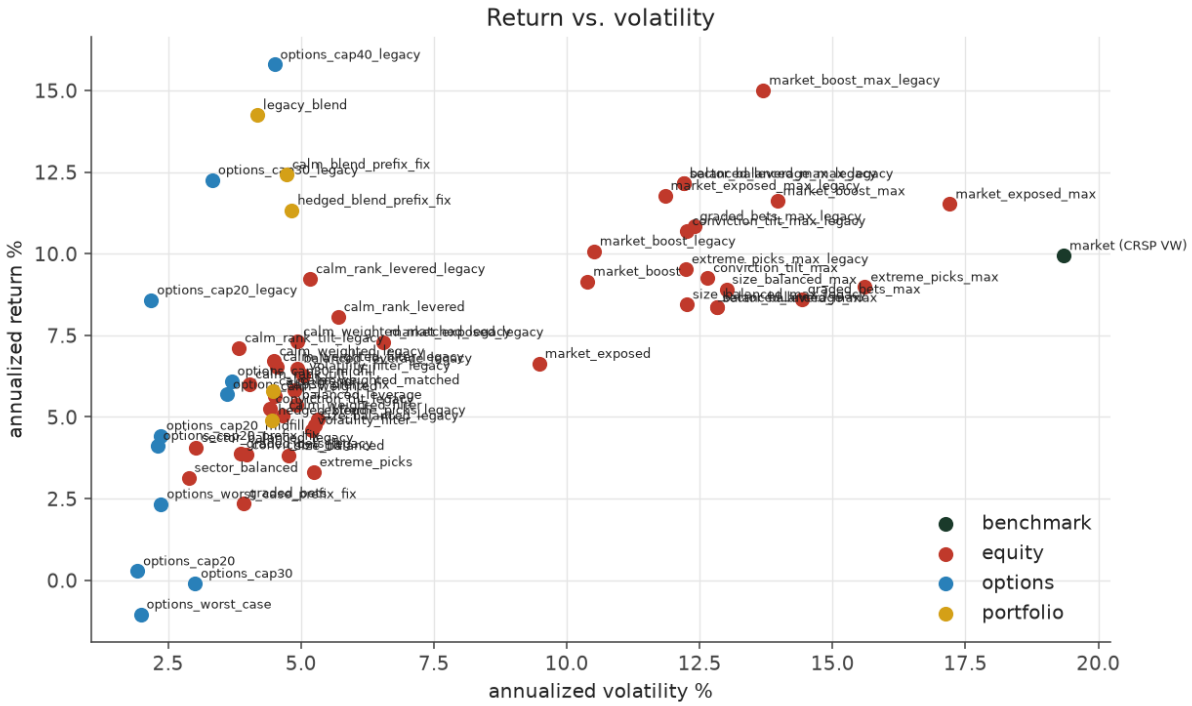
60 strategies, 1996-2013, $10M start, NAV-percentage Sharpe

1. League table

Every strategy that has been run, by asset class and Sharpe. `validated` strategies are candidates for use, `research` are explorations/variants, `reference` are kept as documented counter-examples.

name	family	status	ann. return	ann. vol	sharpe	max DD	calmar	Sharpe 96-01	Sharpe 02-07	Sharpe 08-13
calm_rank_tilt_legacy	equity-volatility	reference	7.1%	3.8%	1.81	-11.6%	0.61	2.94	1.89	0.78
calm_rank_levered_legacy	equity-volatility	reference	9.2%	5.2%	1.73	-15.9%	0.58	2.86	1.96	0.50
calm_weighted_legacy	equity-volatility	reference	6.7%	4.5%	1.47	-15.6%	0.43	2.40	1.63	0.59
calm_rank_tilt	equity-volatility	research	6.0%	4.0%	1.47	-13.6%	0.44	2.31	1.63	0.51
calm_weighted_matched_legacy	equity-volatility	reference	7.3%	4.9%	1.45	-17.0%	0.43	2.39	1.66	0.52
calm_weighted_filter_legacy	equity-volatility	reference	6.5%	4.6%	1.41	-15.7%	0.42	2.26	1.57	0.60
calm_rank_levered	equity-volatility	research	8.1%	5.7%	1.39	-19.6%	0.41	2.17	1.66	0.38
sector_balanced_legacy	equity-neutralized	reference	4.1%	3.0%	1.33	-10.8%	0.37	2.19	1.34	0.66
balanced_leverage_legacy	equity-neutralized	reference	6.5%	4.9%	1.30	-16.6%	0.39	2.07	1.52	0.47
calm_weighted	equity-volatility	validated	5.6%	4.5%	1.24	-17.5%	0.32	2.03	1.54	0.29
volatility_filter_legacy	equity-volatility	reference	6.3%	5.0%	1.23	-16.7%	0.38	1.90	1.44	0.48
calm_weighted_matched	equity-volatility	validated	5.8%	4.9%	1.18	-18.7%	0.31	1.94	1.47	0.25
conviction_tilt_legacy	equity-neutralized	reference	5.2%	4.4%	1.18	-13.1%	0.40	1.74	1.31	0.60
market_exposed_legacy	equity-directional	reference	7.3%	6.6%	1.11	-15.5%	0.47	1.79	1.48	0.32
balanced_leverage	equity-neutralized	validated	5.4%	4.9%	1.09	-18.6%	0.29	1.76	1.45	0.19
market_boost_max_legacy	equity-max	reference	15.0%	13.7%	1.09	-37.0%	0.41	1.39	1.73	0.21
sector_balanced	equity-neutralized	validated	3.1%	2.9%	1.08	-11.4%	0.28	1.72	1.25	0.41
calm_weighted_filter	equity-volatility	research	5.1%	4.7%	1.08	-17.6%	0.29	1.79	1.22	0.30
graded_bets_legacy	equity-baseline	reference	3.9%	3.9%	1.01	-14.0%	0.28	1.39	1.55	0.32
balanced_leverage_max_legacy	equity-max	reference	12.1%	12.2%	1.00	-36.5%	0.33	1.53	1.71	-0.07
sector_balanced_max_legacy	equity-max	reference	12.1%	12.2%	1.00	-36.5%	0.33	1.53	1.71	-0.07
market_exposed_max_legacy	equity-max	reference	11.8%	11.8%	1.00	-33.0%	0.36	1.64	1.65	0.13
conviction_tilt	equity-neutralized	validated	3.8%	4.0%	0.97	-14.5%	0.26	1.24	1.35	0.41
market_boost_legacy	equity-overlay	reference	10.1%	10.5%	0.97	-23.2%	0.43	1.30	1.14	0.48
extreme_picks_legacy	equity-baseline	reference	4.9%	5.3%	0.93	-18.5%	0.27	1.28	1.58	0.28
size_balanced_legacy	equity-baseline	reference	4.7%	5.3%	0.91	-17.5%	0.27	1.32	1.28	0.34
market_boost	equity-overlay	validated	9.2%	10.4%	0.90	-25.1%	0.36	1.20	1.10	0.40
graded_bets_max_legacy	equity-max	reference	10.8%	12.4%	0.89	-38.8%	0.28	1.24	1.63	0.09

conviction_tilt_max_legacy	equity-max	reference	10.7%	12.3%	0.89	-40.0%	0.27	1.30	1.54	0.07
volatility_filter	equity-volatility	research	4.6%	5.2%	0.89	-18.8%	0.25	1.43	1.07	0.19
market_boost_max	equity-max	research	11.6%	14.0%	0.86	-44.6%	0.26	1.17	1.48	-0.05
size_balanced	equity-baseline	validated	3.8%	4.8%	0.81	-18.1%	0.21	0.93	1.44	0.24
extreme_picks_max_legacy	equity-max	reference	9.5%	12.2%	0.80	-36.2%	0.26	1.02	1.72	-0.01
conviction_tilt_max	equity-max	research	9.3%	12.6%	0.76	-40.5%	0.23	1.08	1.45	-0.05
market_exposed	equity-directional	reference	6.6%	9.5%	0.72	-34.1%	0.19	1.42	1.10	0.04
size_balanced_max_legacy	equity-max	reference	8.4%	12.3%	0.72	-37.5%	0.23	1.00	1.52	0.03
size_balanced_max	equity-max	research	8.9%	13.0%	0.72	-43.6%	0.20	0.82	1.52	0.09
market_exposed_max	equity-max	research	11.5%	17.2%	0.72	-53.8%	0.21	1.33	1.13	0.06
balanced_leverage_max	equity-max	research	8.3%	12.8%	0.69	-44.9%	0.19	1.20	1.39	-0.41
sector_balanced_max	equity-max	research	8.3%	12.8%	0.69	-44.9%	0.19	1.20	1.39	-0.41
extreme_picks	equity-baseline	validated	3.3%	5.2%	0.65	-25.2%	0.13	0.86	1.62	-0.11
graded_bets_max	equity-max	research	8.6%	14.4%	0.64	-48.0%	0.18	0.97	1.53	-0.20
extreme_picks_max	equity-max	research	9.0%	15.6%	0.63	-56.4%	0.16	0.78	1.58	-0.10
graded_bets	equity-baseline	validated	2.4%	3.9%	0.62	-19.5%	0.12	0.96	1.52	-0.23
options_cap20_legacy	options-kelly	reference	8.6%	2.2%	3.78	-10.3%	0.83	1.64	4.80	4.39
options_cap30_legacy	options-kelly	reference	12.3%	3.3%	3.49	-17.4%	0.70	0.98	4.67	4.10
options_cap40_legacy	options-kelly	reference	15.8%	4.5%	3.28	-23.9%	0.66	0.54	4.58	3.93
options_cap20_midfill	options-kelly	reference	4.4%	2.4%	1.85	-14.5%	0.31	1.19	2.01	2.37
options_cap20_prefix_fix	options-kelly	reference	4.1%	2.3%	1.77	-13.8%	0.30	0.69	2.04	2.39
options_cap30_midfill	options-kelly	reference	6.1%	3.7%	1.62	-22.5%	0.27	0.57	1.91	2.16
options_cap30_prefix_fix	options-kelly	reference	5.7%	3.6%	1.55	-22.3%	0.26	0.12	1.95	2.17
options_worst_case_prefix_fix	options-kelly	reference	2.3%	2.4%	0.99	-17.1%	0.14	0.32	0.99	1.60
options_cap20	options-kelly	validated	0.3%	1.9%	0.16	-18.9%	0.01	0.07	0.31	0.00
options_cap30	options-kelly	validated	-0.1%	3.0%	-0.02	-29.0%	-0.00	-0.48	0.28	-0.10
options_worst_case	options-kelly	research	-1.0%	2.0%	-0.52	-32.3%	-0.03	-0.30	-0.58	-0.66
legacy_blend	portfolio	reference	14.2%	4.2%	3.21	-9.9%	1.44	2.79	4.64	2.61
calm_blend_prefix_fix	portfolio	reference	12.4%	4.7%	2.50	-15.5%	0.80	2.92	2.84	2.10
hedged_blend_prefix_fix	portfolio	reference	11.3%	4.8%	2.25	-16.7%	0.68	2.41	2.70	1.91
calm_blend	portfolio	research	5.8%	4.5%	1.28	-20.1%	0.29	2.13	1.36	0.43
hedged_blend	portfolio	validated	4.9%	4.5%	1.09	-21.8%	0.22	1.82	1.31	0.25



Each dot is one backtest. Options books and the hedged blend sit high-left (high return per unit of risk).

2. What each iteration changed

strategy	parent	d return	d vol	d sharpe	d maxdd
balanced_leverage	sector_balanced	+2.24%	+2.02%	+0.01	-7.2%
balanced_leverage_legacy	balanced_leverage	+1.12%	+0.03%	+0.21	+1.9%
balanced_leverage_max	balanced_leverage	+2.97%	+7.91%	-0.40	-26.3%
balanced_leverage_max_legacy	balanced_leverage_max	+3.80%	-0.62%	+0.31	+8.4%
calm_blend	hedged_blend	+0.90%	+0.01%	+0.19	+1.7%
calm_blend_prefix_fix	calm_blend	+6.63%	+0.25%	+1.22	+4.6%
calm_rank_levered	calm_rank_tilt	+2.06%	+1.66%	-0.08	-6.0%
calm_rank_levered_legacy	calm_rank_levered	+1.16%	-0.53%	+0.34	+3.7%
calm_rank_tilt	calm_weighted	+0.37%	-0.47%	+0.23	+3.9%
calm_rank_tilt_legacy	calm_rank_tilt	+1.11%	-0.20%	+0.35	+2.0%
calm_weighted	balanced_leverage	+0.26%	-0.40%	+0.15	+1.0%
calm_weighted_filter	balanced_leverage	-0.32%	-0.24%	-0.01	+0.9%
calm_weighted_filter_legacy	calm_weighted_filter	+1.48%	-0.11%	+0.33	+2.0%
calm_weighted_legacy	calm_weighted	+1.07%	-0.02%	+0.23	+1.9%
calm_weighted_matched	calm_weighted	+0.19%	+0.37%	-0.05	-1.1%
calm_weighted_matched_legacy	calm_weighted_matched	+1.48%	+0.05%	+0.27	+1.6%
conviction_tilt	graded_bets	+1.47%	+0.06%	+0.35	+5.0%

conviction_tilt_legacy	conviction_tilt	+1.41%	+0.44%	+0.21	+1.4%
conviction_tilt_max	conviction_tilt	+5.42%	+8.68%	-0.20	-26.0%
conviction_tilt_max_legacy	conviction_tilt_max	+1.43%	-0.40%	+0.13	+0.5%
extreme_picks_legacy	extreme_picks	+1.59%	+0.08%	+0.28	+6.7%
extreme_picks_max	extreme_picks	+5.67%	+10.37%	-0.02	-31.3%
extreme_picks_max_legacy	extreme_picks_max	+0.55%	-3.37%	+0.18	+20.2%
graded_bets_legacy	graded_bets	+1.52%	-0.05%	+0.39	+5.5%
graded_bets_max	graded_bets	+6.24%	+10.51%	+0.03	-28.6%
graded_bets_max_legacy	graded_bets_max	+2.23%	-2.03%	+0.25	+9.3%
hedged_blend	calm_weighted	-0.74%	-0.05%	-0.14	-4.3%
hedged_blend_prefix_fix	hedged_blend	+6.43%	+0.37%	+1.15	+5.1%
legacy_blend	hedged_blend	+9.35%	-0.28%	+2.11	+11.9%
market_boost	balanced_leverage	+3.77%	+5.48%	-0.20	-6.5%
market_boost_legacy	market_boost	+0.93%	+0.12%	+0.07	+1.9%
market_boost_max	balanced_leverage_max	+3.27%	+1.14%	+0.17	+0.3%
market_boost_max_legacy	market_boost_max	+3.39%	-0.26%	+0.23	+7.7%
market_exposed	balanced_leverage	+1.26%	+4.57%	-0.37	-15.6%
market_exposed_legacy	market_exposed	+0.66%	-2.93%	+0.38	+18.6%
market_exposed_max	market_exposed	+4.89%	+7.72%	-0.01	-19.6%
market_exposed_max_legacy	market_exposed_max	+0.26%	-5.36%	+0.28	+20.8%
options_cap20_midfill	options_cap20	+4.15%	+0.43%	+1.69	+4.4%
options_cap20_prefix_fix	options_cap20	+3.84%	+0.38%	+1.61	+5.0%
options_cap30	options_cap20	-0.38%	+1.08%	-0.17	-10.1%
options_cap30_legacy	options_cap20_legacy	+3.68%	+1.15%	-0.29	-7.1%
options_cap30_midfill	options_cap20_midfill	+1.67%	+1.34%	-0.23	-8.1%
options_cap30_prefix_fix	options_cap30	+5.79%	+0.60%	+1.57	+6.7%
options_cap40_legacy	options_cap20_legacy	+7.21%	+2.32%	-0.50	-13.6%
options_worst_case	options_cap20	-1.32%	+0.06%	-0.67	-13.5%
options_worst_case_prefix_fix	options_worst_case	+3.38%	+0.38%	+1.50	+15.2%
sector_balanced	conviction_tilt	-0.70%	-1.08%	+0.12	+3.1%
sector_balanced_legacy	sector_balanced	+0.92%	+0.13%	+0.25	+0.6%
sector_balanced_max	sector_balanced	+5.21%	+9.93%	-0.40	-33.5%
sector_balanced_max_legacy	sector_balanced_max	+3.80%	-0.62%	+0.31	+8.4%
size_balanced	extreme_picks	+0.50%	-0.47%	+0.16	+7.1%
size_balanced_legacy	size_balanced	+0.92%	+0.49%	+0.10	+0.5%

size_balanced_max	size_balanced	+5.09%	+8.23%	-0.09	-25.6%
size_balanced_max_legacy	size_balanced_max	-0.46%	-0.74%	+0.00	+6.2%
volatility_filter	balanced_leverage	-0.77%	+0.30%	-0.20	-0.2%
volatility_filter_legacy	volatility_filter	+1.66%	-0.16%	+0.34	+2.1%

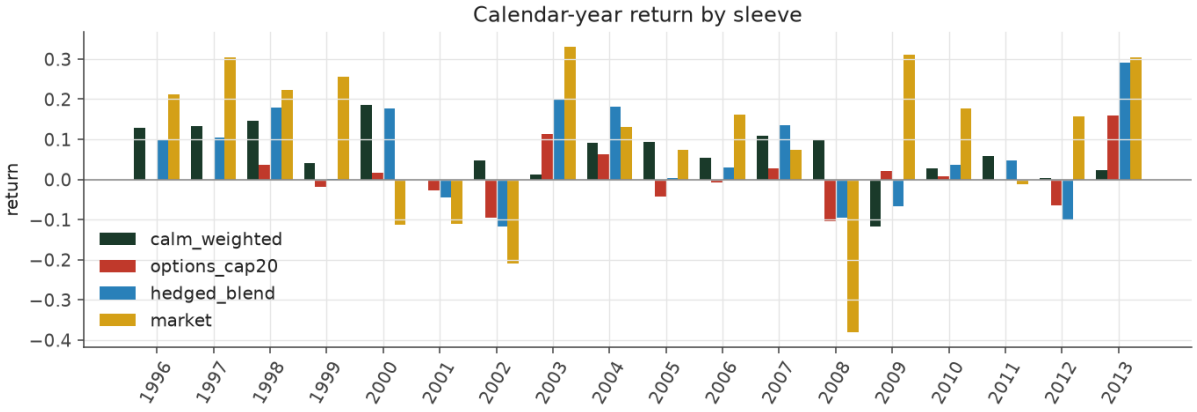
d_* columns are the change from the parent strategy. Reading down: neutralizing size and sector (Conviction Tilt -> Sector Balanced) and inverse-vol sizing (Balanced Leverage -> Calm Weighted) buy Sharpe; raising the leverage cap buys return; the max-size variants buy return at the cost of Sharpe because the liquidity cap starts to bind.

3. Stability across eras

name	Sharpe 96-01	Sharpe 02-07	Sharpe 08-13
extreme_picks	0.86	1.62	-0.11
graded_bets	0.96	1.52	-0.23
conviction_tilt	1.24	1.35	0.41
sector_balanced	1.72	1.25	0.41
balanced_leverage	1.76	1.45	0.19
market_exposed	1.42	1.10	0.04
calm_weighted	2.03	1.54	0.29
volatility_filter	1.43	1.07	0.19
calm_rank_tilt	2.31	1.63	0.51
market_boost	1.20	1.10	0.40
options_cap20	0.07	0.31	0.00
options_cap30	-0.48	0.28	-0.10
options_cap20_legacy	1.64	4.80	4.39
hedged_blend	1.82	1.31	0.25

year	calm weighted	options cap20	hedged blend	market
1996	12.8%	0.0%	10.1%	21.2%
1997	13.3%	0.0%	10.5%	30.4%
1998	14.6%	3.6%	18.0%	22.3%
1999	4.0%	-1.9%	0.1%	25.7%
2000	18.6%	1.6%	17.7%	-11.3%
2001	-0.2%	-2.8%	-4.5%	-11.0%
2002	4.7%	-9.6%	-11.8%	-20.9%
2003	1.3%	11.3%	20.0%	33.1%
2004	9.1%	6.3%	18.2%	13.0%
2005	9.4%	-4.2%	0.4%	7.3%

2006	5.3%	-0.8%	2.9%	16.2%
2007	10.9%	2.7%	13.4%	7.3%
2008	9.7%	-10.3%	-9.5%	-38.2%
2009	-11.8%	2.0%	-6.6%	31.2%
2010	2.7%	0.8%	3.6%	17.7%
2011	5.9%	-	4.7%	-1.1%
2012	0.3%	-6.6%	-10.1%	15.8%
2013	2.3%	15.9%	29.0%	30.5%



Equity alpha is front-loaded (1996-2003); the options book is back-loaded (2003-2013) and lost in 2008 when equity gained.

The equity drift has decayed: Calm Weighted's return fell from ~14% a year in 1996-99 to ~2-5% in 2011-13, the usual signature of PEAD being arbitraged. The options book shows the opposite shape in this sample, which is part of why the two combine well -- and part of why its recent strength should not be extrapolated without a skeptical look.

4. Equity books: long leg vs. short leg

strategy	long pnl M	short pnl M	long excess M	short excess M	long bp per dollar	short bp per dollar	corr long short
extreme_picks	18.3	-6.7	7.6	4.1	259.7	-110.4	-0.59
graded_bets	15.1	-6.7	4.8	4.0	223.4	-107.4	-0.71
conviction_tilt	20.5	-6.3	8.8	6.1	293.2	-80.2	-0.77
sector_balanced	16.2	-5.4	7.2	3.8	277.9	-90.8	-0.79
balanced_leverage	36.7	-13.1	13.3	11.0	252.7	-88.0	-0.80
calm_weighted	37.5	-12.3	13.6	12.2	252.2	-81.1	-0.79

• **In every equity strategy the short leg loses money in absolute terms** while the long leg earns several times the total. The headline is real but misleading: returns are measured against the cap-weighted market, and the average event stock beats that benchmark (small-cap tilt plus the earnings-announcement premium), flattering the long leg and penalizing the short leg by the same amount.

• Stripped of that common component, **the surprise signal earns the same on both legs** (columns `*_excess_M`): the shorts are pulling their weight, they are just paid in relative performance, not in absolute

declines.

• The legs' daily P&L are ~-0.8 correlated: the short leg is a hedge. Dropping it for a long-only book would multiply the volatility and add market-relative factor exposure without adding alpha.

• The backtest charges no stock-borrow fee. The short leg is small-cap-heavy where borrow is expensive; each 1% a year of borrow costs ~0.6pp of Balanced Leverage's return (see the per-strategy report's borrow table).

• Alpha per dollar traded is ~7x higher in the smallest size quintile (~11x in the least liquid quintile) than in the largest, and the >90-day holding cells earn ~8x the alpha per dollar of the <=40-day cells. The book's capacity is therefore bounded by small illiquid names.

5. Options book: what the return is made of, and how much of it is real

The originally reported options result does not survive a careful look at the quotes. The original backtest marked a missing forward quote at the previous live quote. 22% of picks have no quote at 60 days and 93% of those are zero-bid options -- dead contracts nobody will buy -- which that convention never marked down. Corrected: 8.58% / Sharpe 3.78 as reported -> 0.28% / 0.16 (dead quotes at half-ask, plus a $0.05 minimum tick cost) -> -1.04% / -0.52 if dead options are worth nothing. A real, smaller edge remains.

• 97.1% of funded trades are calls, producing 59.8% of gross P&L -- across ALL deciles, including the worst surprises. The Kelly selector finds cheap OTM calls better than puts even on bad news.

• Win rate 31%, median trade -50%, mean 1%: a convex pay-off. The top 1% of trades are 1151% of gross P&L.

• What survives the corrected marks is concentrated in calls struck at or just out of the money (1.0-1.2x spot) on low-volatility stocks. Deep in-the-money calls absorb about half the premium budget and earn ~5% on it; high-volatility stocks earn ~7%.

• Return on premium does not rise with SUE decile, so the edge is not the directional drift alone: it is mostly cheap post-earnings convexity on calm stocks, in a sample that was net bullish and whose profit is dominated by two years (2003 and 2013).

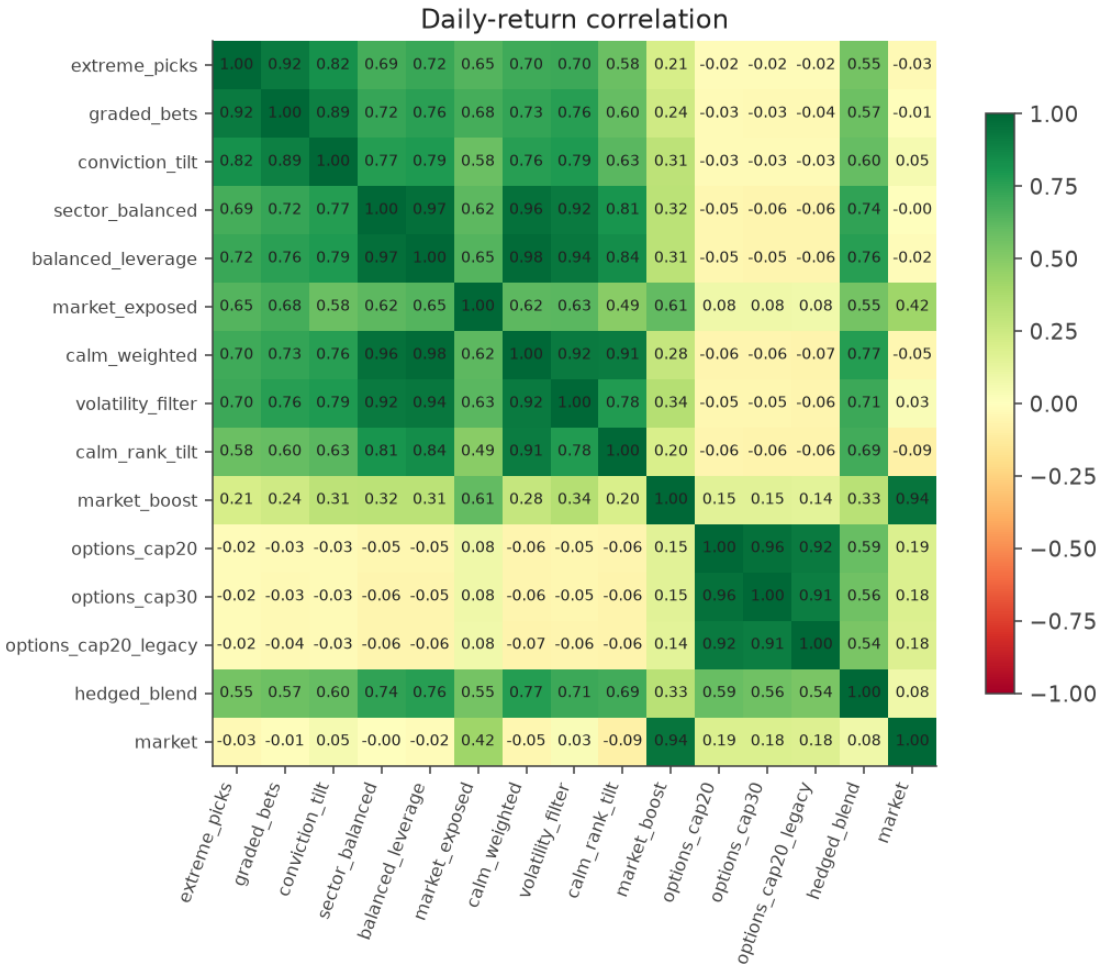
vol tercile	trades	premium M	gross M	net M	share of net	roi on premium
low vol	19,712	14.15	1.25	1.25	258.4%	8.8%
mid vol	19,712	12.36	0.04	0.04	8.9%	0.3%
high vol	19,712	20.38	-0.81	-0.81	-167.6%	-4.0%

Execution and marking sensitivity

case	ann. return	ann. vol	sharpe	max DD	trades
as registered	0.28%	1.92%	0.16	-18.9%	59,153
ORIGINAL convention: dead quotes carried forward, flat 3% cost	2.45%	1.82%	1.33	-12.9%	62,494
dead quotes at half-ask, flat 3% cost (no tick floor)	0.28%	1.92%	0.16	-18.9%	59,153
dead quotes worth zero (pessimistic bound)	-1.04%	1.99%	-0.52	-32.3%	57,102
+ $0.10/share minimum cost	-3.28%	1.99%	-1.66	-52.3%	53,826
skip contracts under $0.25	0.03%	1.73%	0.03	-18.6%	47,184
skip contracts under $0.50	-0.06%	1.60%	-0.03	-19.1%	39,309
flat 10% round-trip cost	-2.31%	1.96%	-1.19	-44.4%	54,475

Treat the corrected `options_cap20` / `options_cap30` as the options sleeve and the `_legacy` strategies as the historical headline only. Everything the hedged blend says about options is built on the corrected numbers.

6. How the strategies move together



Equity books cluster together; options are independent of them.

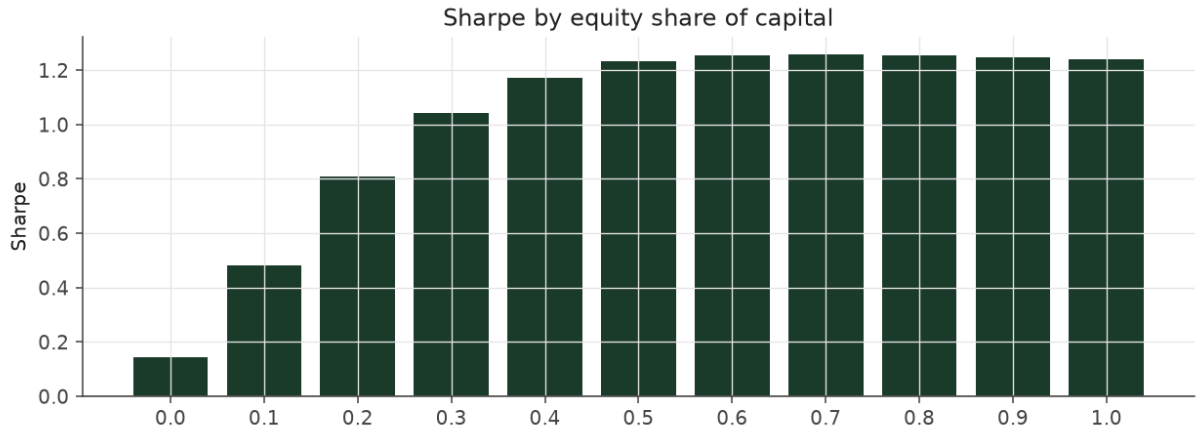
• All market-neutral equity variants are 0.7-0.99 correlated -- they are one bet expressed several ways, so combining them adds nothing.

• Equity vs. options correlation: -0.06. The options book's correlation to the market is 0.19; the market-neutral equity books' is ~0.

• `market_boost` is 0.94 correlated with the market by construction; it answers 'beat the S&P', not 'diversify'.

7. Combining an equity book with an options book

Capital split: fraction of NAV in each sleeve (weights sum to 1)



Sharpe peaks around 15-25% equity: diversification, then the better sleeve dominates.

method	w equity	w options	ann. return	ann. vol	sharpe	max DD
equal	0.50	0.50	2.9%	2.4%	1.23	-9.0%
inverse_vol	0.28	0.72	1.8%	1.8%	1.01	-10.2%
max_sharpe (in-sample)	0.69	0.31	4.0%	3.1%	1.26	-11.8%
walk-forward max-Sharpe (out-of-sample)	0.63	0.37	3.2%	3.0%	1.09	-12.9%

Stacked on one NAV (weights are multiples of each sleeve's own sizing)

target vol	equity multiple	options multiple	ann. return	ann. vol	sharpe	max DD	worst day
3%	0.49	1.23	3.0%	3.0%	1.01	-16.9%	-1.8%
5%	0.81	2.05	5.0%	5.0%	1.01	-26.6%	-3.0%
8%	1.29	3.28	8.1%	8.0%	1.01	-39.2%	-4.7%

Equal-risk weights (each sleeve contributes the same stand-alone volatility). The 5% row is the registered `hedged_blend`. Larger multiples assume the extra options premium and equity gross exposure can actually be financed and traded at the backtested costs.

The hedge in action: stress years

year	calm weighted	options cap20	hedged blend	market
2000	18.6%	1.6%	17.7%	-11.3%
2001	-0.2%	-2.8%	-4.5%	-11.0%
2002	4.7%	-9.6%	-11.8%	-20.9%
2008	9.7%	-10.3%	-9.5%	-38.2%
2009	-11.8%	2.0%	-6.6%	31.2%

• The options book's worst year (2008, -10.3%) coincided with the equity book's +9.7%; in 2009 the signs flipped. That offset is the hedge -- it comes from low correlation across time, not from any offsetting position.

• The blend's market beta is 0.022: there is nothing left for an index hedge to remove. Layering a short-index overlay on top would only pay costs; a market-beta overlay (`market_boost`) goes the other way, adding the equity risk premium on purpose.

• Everything here is in-sample except the walk-forward row. The caveats that apply to the sleeves apply to the blend: the options sleeve's costs, the equity sleeve's missing borrow fees, and no financing cost on the stacked multiples.

8. Conclusions and next steps

• Use `calm_weighted` as the equity alpha sleeve (validated, no fitted parameters, ahead of Balanced Leverage in every era; `calm_rank_tilt` is the stronger research candidate) and `options_cap20` (or the cap30 variant) as the options sleeve -- a real but modest edge once dead quotes are marked down -- and blend by equal risk (`hedged_blend`).

• Do not add more market-neutral equity variants -- they are one bet. New equity ideas must change the signal or the holding rule, not the sizing.

• Options: the quote convention was the biggest error found in this project. Next: backtest with real bid/ask on entry and exit (the data has them), restrict to the contract region that survives (0.97-1.2x moneyness, low vol, premium >= $0.25), and test on a holdout -- 2003 and 2013 carry most of the profit.

• The measured weak spots to attack on the equity side: the era decay (alpha per dollar fell ~4x), the <=40-day holding cells that earn little after costs, and the mid deciles (D7, D9) that earn almost no excess per dollar.